PROJECT OVERVIEW:

This project analyzes multi-year E-commerce transactional data to evaluate overall business performance, profitability, marketing efficiency, and regional contribution. The objective was to transform raw transactional data into meaningful business insights using PostgreSQL for advanced SQL analysis and Python for visualization.

The dataset spans four years (2021–2024) and includes information on revenue, profit, advertising spend, product categories, units sold, and regional performance.

2. Data Preparation & Methodology

The workflow included:

- Data cleaning and preprocessing using Python (Pandas)
- Loading structured data into PostgreSQL using SQLAlchemy
- Writing advanced SQL queries using:
    - Aggregations
    - Common Table Expressions (CTEs)
    - Window functions (LAG)
    - Type casting and safe division

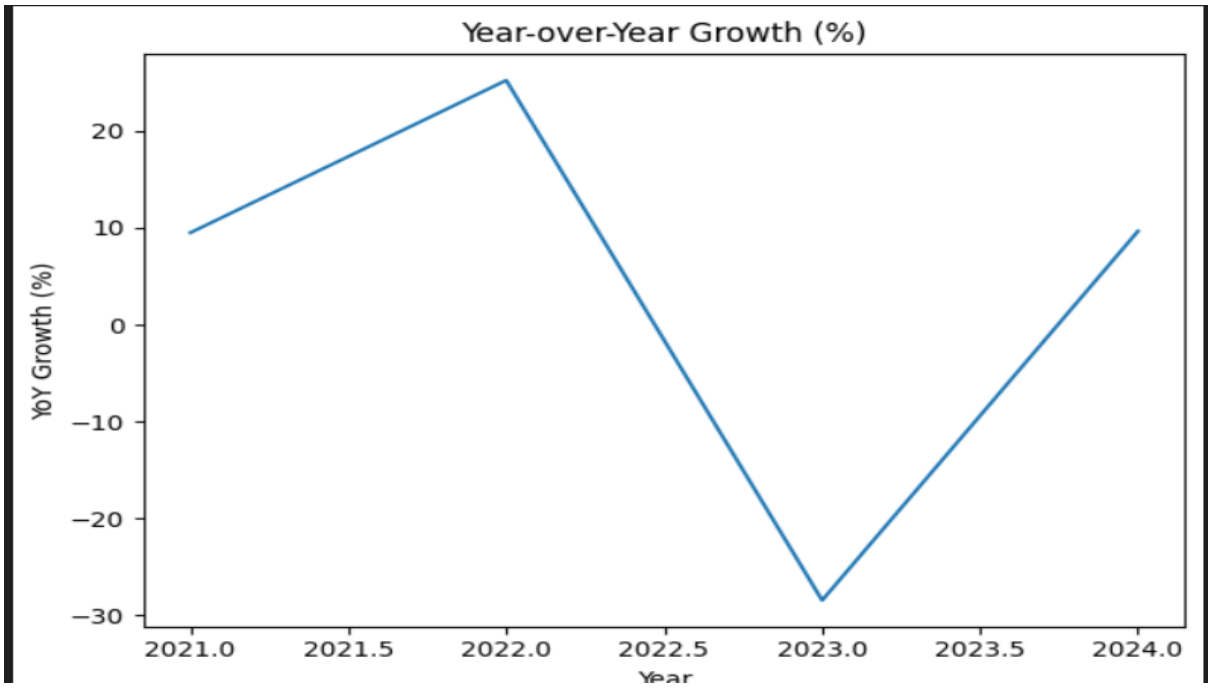
- Pulling analytical outputs into Python
- Creating visualizations using matplotlib

Key performance indicators (KPIs) calculated:

- Total Revenue
- Profit Margin (%)
- Year-over-Year (YoY) Growth
- Return on Ad Spend (ROAS)
- Region-wise Revenue Contribution

VISUALIZATIONS WITH INSIGHTS:

1. Year-over-Year (YoY) Growth Analysis:

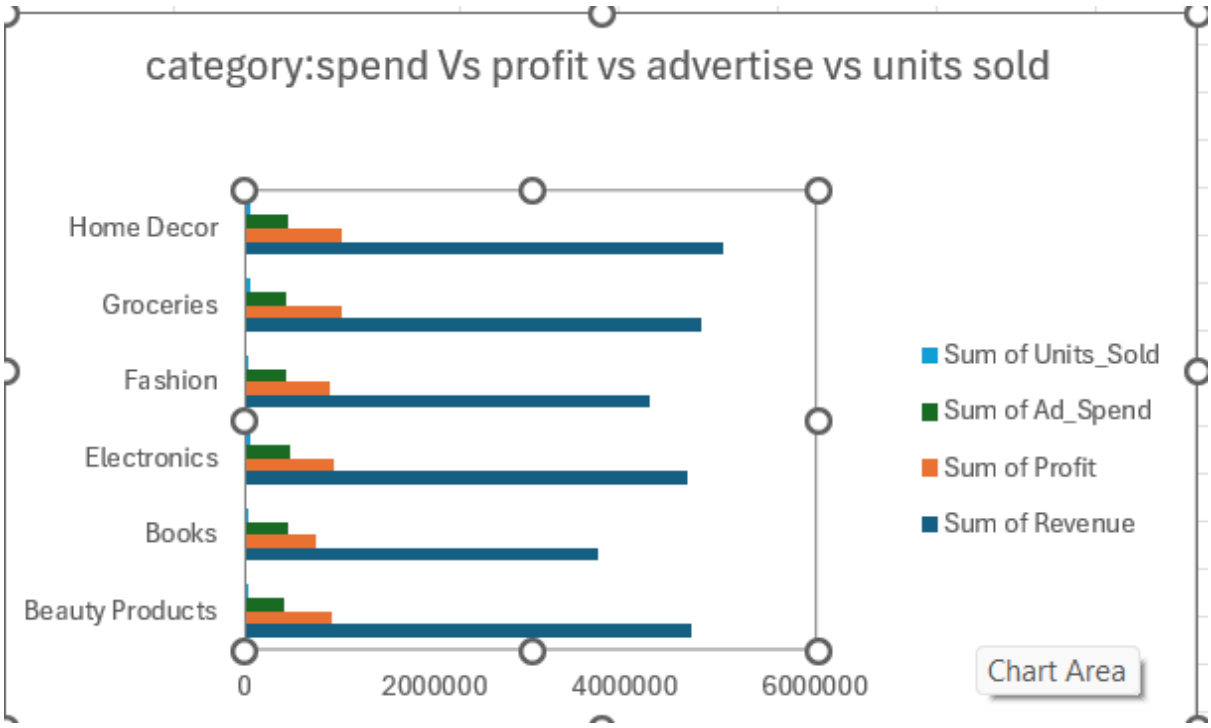


Insight:

The company experienced strong expansion until 2022, then sharp downturn in 2023. This could indicate external market slowdown, operational challenges.

However, the positive recovery in 2024 demonstrates business resilience and adaptability.

2. Category-Level Performance Analysis:



Insight:

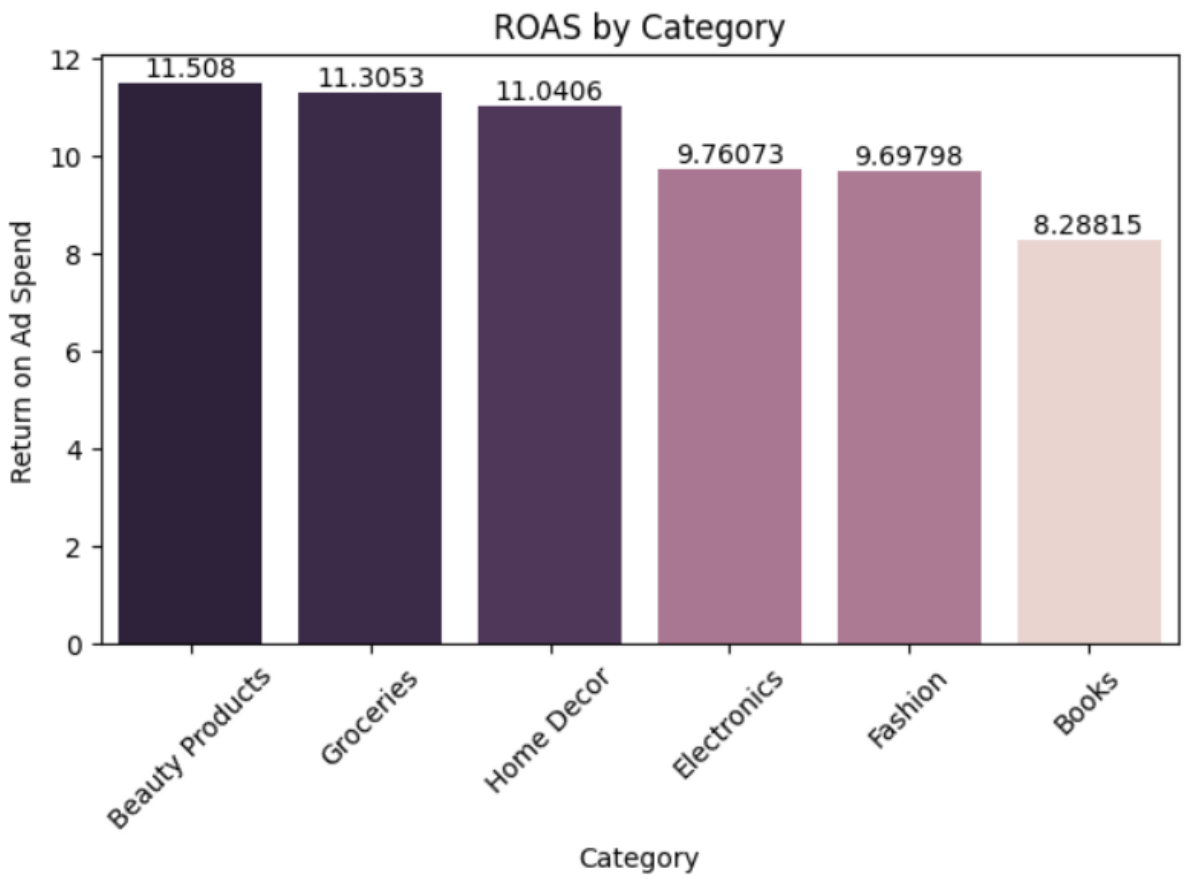
This visualization highlights that the company maintains a well-balanced category portfolio. Revenue and profit remain proportionally aligned, indicating controlled cost management.

However, advertising efficiency varies slightly across categories, particularly for Books, suggesting opportunities for marketing optimization.

High-revenue categories such as Home Decor and Electronics likely benefit from higher average selling prices, while Groceries and Beauty operate more on volume efficiency.

Overall, the business demonstrates strong operational consistency, with selective opportunities to enhance marketing ROI and category-level profitability.

3. Marketing Performance (ROAS Analysis):

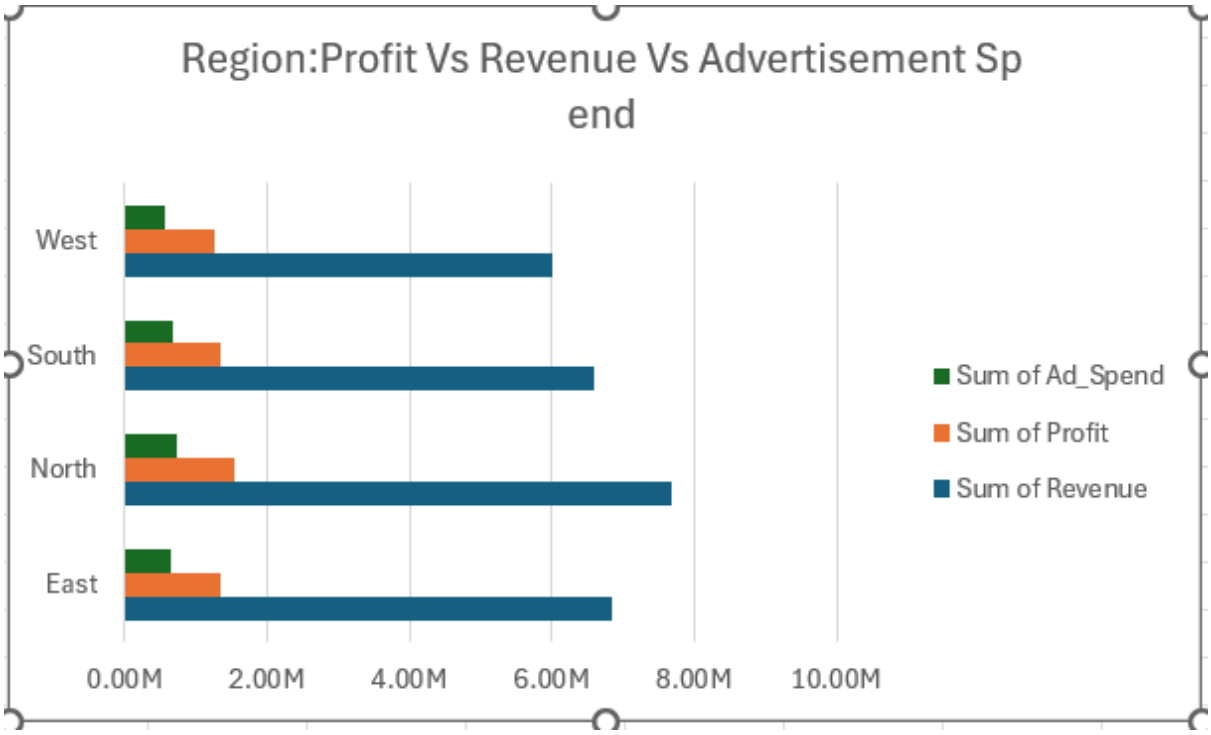


Insight:

Beauty Products and Groceries generate the highest return per advertising dollar spent. This means marketing campaigns in these categories are highly effective.

Books shows weaker marketing efficiency and may require campaign optimization or budget reassessment.

4. Regional Performance Analysis:

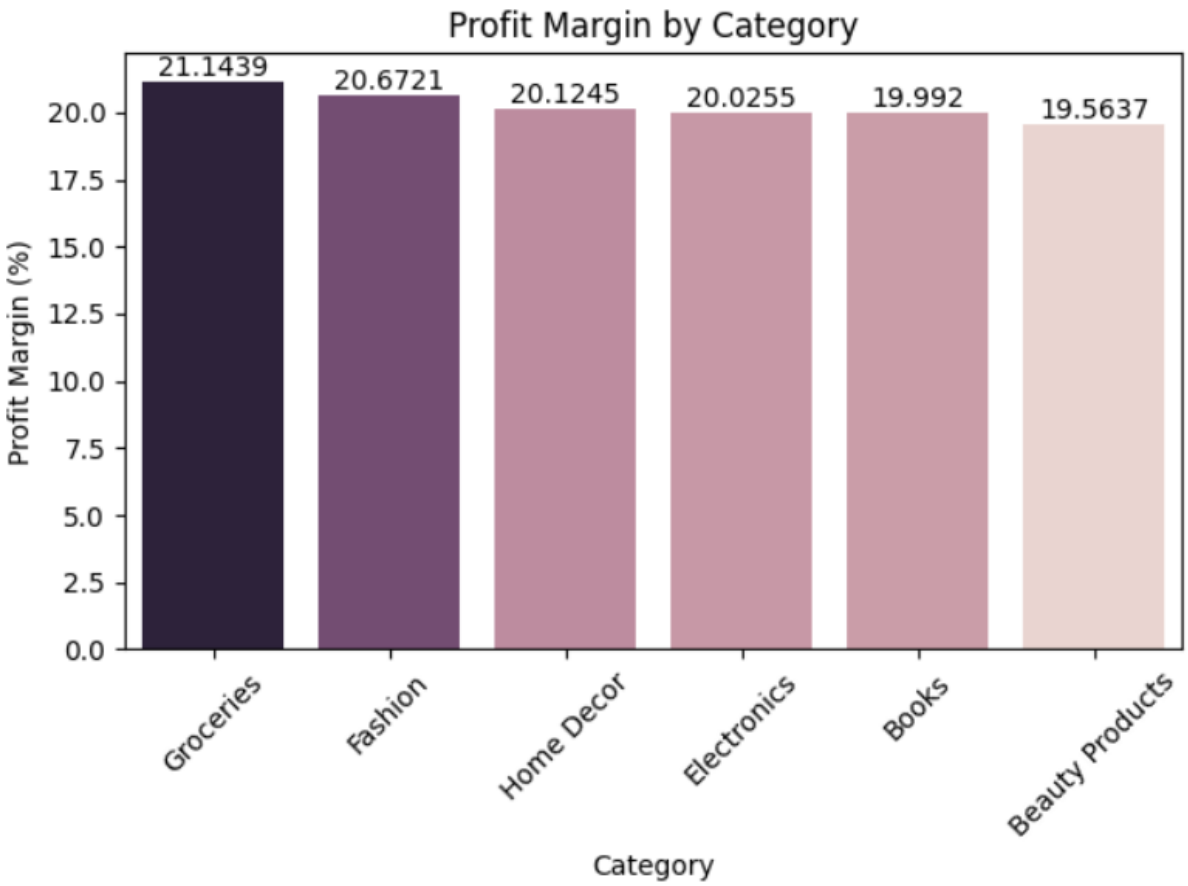


Insight:

North region serves as the primary revenue engine for the company.

West region shows growth potential and may benefit from expansion strategies.

5. Profit Margin by Category:



Insight: Profit margin variation is minimal.

This indicates:

- Strong cost control across categories
- Consistent pricing strategy
- Balanced cost structure

RECOMMENDATION:

The company demonstrates stable and well-distributed category performance. However, marketing budget optimization and category-level strategic focus can significantly improve profitability. Investment should be increased in high-performing categories such as Home Decor and Beauty Products, while underperforming segments like Books require restructuring or reduced marketing allocation. A data-driven allocation strategy will enhance return on ad spend and support sustainable long-term growth.

SUMMARY:

This end-to-end E-commerce performance analysis highlights a resilient and diversified business model. While revenue experienced

significant volatility between 2022 and 2023, profitability remains stable across categories. Marketing efficiency is strongest in Beauty Products and Groceries, indicating opportunities for strategic budget reallocation.

Home Decor stands as the primary revenue driver, while Groceries leads in margin efficiency. Regional analysis shows North as the dominant market, with West offering expansion potential.

Strategic focus areas should include:

- Stabilizing growth volatility
- Increasing investment in high-ROAS categories
- Optimizing Electronics marketing efficiency
- Expanding regional footprint in underperforming markets

Overall, the company maintains strong operational health with clear opportunities for scalable and sustainable growth.